SHUVRO AHMED

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EDUCATION HISTORY

Department of Computer and Electrical Engineering, Mid Sweden University — **Sundsvall**
Master of Science — *Aug 2026 – Ongoing*
- Current Grade: N/A
- Credits Completed: N/A
- 2nd Year of Erasmus Mundus Joint Masters in Imaging (Study Track – Visual AI)

Department of Electronics, Information and Bioengineering, Politecnico di Milano — **Milan**
Master of Science — *Sep 2025 – Ongoing*
- Current Grade: 25.67/30 (18 = Pass, 30 = Highest)
- Credits Completed: 45/60 (ECTS)
- 1st Year of Erasmus Mundus Joint Masters in Imaging (Study Track – Visual AI)

Department of Computer Science and Engineering, BRAC University — **Dhaka**
Bachelor of Science — *Sep 2018 – Mar 2023*
- CGPA: 3.77
- Dean's List & VC's List: 5 Times
- Academic Distinction: High Distinction

PROFESSIONAL EXPERIENCE

BRAC Business School, BRAC University — **Dhaka**
Teaching Assistant — *Oct 2023 – Ongoing*
- Conducted tutorial class on CSE101 (Introduction to Computer Science) & STA101 (Introduction to Statistics)
- Analyzed business and migration data using Python, identifying food security as the top migration driver (29.9%)

ITIL — **Dhaka**
AI Engineer Intern — *Aug 2023 – Dec 2023*
- Participated in training workshops related to natural language processing & machine learning funded by the World Bank
- Designed and executed various unsupervised learning techniques such as K means clustering

Information Technology, Aarong — **Dhaka**
Software Engineer Intern — *May 2023 – Aug 2023*
- Designed and developed an AI-based chatbot to streamline customer service and enhance user engagement
- Proposed and designed a queue management system using Figma & YOLO algorithm

Mathematics and Natural Sciences Department, BRAC University — **Dhaka**
Student Tutor — *May 2022 – Dec 2022*
- Provided consultation for PHY101 (Introduction to Physics) & PHY111 (Principles of Physics I) to over 160 students
- Graded assignments for over 160 students

RESEARCH

Research Profile - sciprofiles.com/profile/3009726

Conference proceedings

i) Ahmed, S., Firoz, N., Tahsin, M. S., & Akpuluma, D. (2025, June). Forecasting Carbon Footprint: Temporal Fusion Transformer Modeling for Predicting CO2 Emissions in Bangladesh. In International Conference on Innovations in Cybersecurity and Data Science Proceedings of ICICDS (pp. 387-401). Cham: Springer Nature Switzerland.
- Presented at International Conference on Innovations in Cybersecurity and Data Science

ii) *Rabbi, R., Datta, J., Ahmed, S., & Zereen, A. N. (2024, December). Wildfire Prediction using Convolutional Kolmogorov-Arnold Network. In 2024 13th International Conference on Electrical and Computer Engineering (ICECE) (pp. 667-672). IEEE.*
- Presented at 13th International Conference on Electrical and Computer Engineering (ICECE)

iii) *Ahmed, S., Karmoker, J., Mojumder, R., Rahman, M. M., Alam, M. G. R., & Reza, M. T. (2023). Hyperautomation in super shop using machine learning. Engineering Proceedings, 39(1), 63.*
- Presented at the 9th International Conference on Time Series and Forecasting

Journal Papers

i) A Systematic Review on Machine Learning and Optimization Theories for Smart Retail Management System: Applications, Methods, Challenges, and Future Research Directions
- Manuscript under review at SN Computer Science journal

Works in Progress

i) Improving VQ-VAE-2 for unsupervised reconstruction-based anomaly detection

HONORS & AWARDS

Erasmus Mundus Joint Masters in Imaging — **Milan, Sundsvall, Tampere**
Fully Funded Scholarship — *Sep 2025 - Aug 2027*

BRAC University — **Dhaka**
Merit Scholarship Based on BRAC University Academic Results — *May 2020 - Apr 2022*

PROJECTS

View Projects - github.com/Shuvro-Ahmed/

Image Analysis and Computer Vision — *Sep 2025 – Feb 2026*

i) Single-View 3D Reconstruction of a Ribbed Cylindrical Vault
- *Reconstructed 3D vault geometry from a single image using projective geometry, vanishing points, and geometric constraint*

ii) Visual reconstruction of played music
- *Tracked hand landmarks and finger presses with MediaPipe to map fingertips to piano keys and reconstruct played notes*

Applied Statistics — *Feb 2026 – Sep 2026*

i) Predicting Municipal IFC Using GRINS Dataset
- *Predicted Italy's IFC using panel-aware LASSO and nested region linear mixed-effects models, achieving CV $R^2 = 0.853$*

Advanced Deep Learning — *Feb 2026 – Sep 2026*

i) Pixel-Level Anomaly Segmentation in Multi-View Images
- *Built a U-Net, DeepLabV3+ & SegFormer ensemble with synthetic anomaly generation and TTA, achieving 0.9264 Pixel-AP*

Artificial Neural Network and Deep Learning — *Sep 2025 – Feb 2026*

i) Pain-Level Classification from Multivariate Time Series
- *Combined CNN–BiGRU–Attention temporal modeling with LightGBM global motion features, achieving 0.9427 macro-F1*

ii) Molecular Subtype Classification from Histopathology Images
- *Used a 3 seed EfficientNet-B0 ensemble on image/mask pairs with ROI cropping & TTA,, achieving 0.3784 macro-F1*

Foundation of Operations Research — *Sep 2025 – Feb 2026*

i) Optimization of Drone Route
- *Optimized 4-drone 3D inspection routes via MLP, enforcing exactly once coverage while minimizing drone's return time*

Software Engineering 2 — *Sep 2025 – Feb 2026*

i) Requirement Analysis and Specification Document for Best Bike Paths
- *Specified actors, use cases, functional/non-functional requirements & system constraints for a route management platform*

ii) Design Document for Best Bike Paths
- *Designed the component architecture, interfaces, data model & interaction flows for the Best Bike Paths system*

Bachelor Thesis — *Jan 2022 – Jan 2023*

i) Daily Sales Forecasting in a Supershop
- *Implemented hybrid linear-XGBoost model, achieving a Mean Absolute Percentage Error (MAPE) of 0.09*

ii) Product Placement in a Supershop
- *Implemented the FP-Growth algorithm, generating rules in 0.473 seconds with a minimum support of 0.03*

iii) Voice Bot
- *Developed a voice chatbot using Spacy for tokenization & NLTK's BLEU function for response accuracy*

ACTIVITIES

IEEE International Conference on Image Processing — **Tampere**
Volunteer — *Sep 2026 - Sep 2026*

Buddy Project by Politecnico di Milano — **Milan**
Volunteer — *Feb 2026 - Ongoing*

Robotics Club of BRAC University — **Dhaka**
Junior Executive — *Jan 2020 - Dec 2021*

WORKSHOPS ATTENDED

i) Introduction to Image Signal Processors (ISP) and Camera Control Algorithms (3A+) by IEEE ICIP 2026
ii) EDGE–AI Training Program by IIT, University of Dhaka
iii) Basics of Robotics by ROBU
iv) Basics of Animated Film Making by Charuputhi

SKILLS & INTERESTS

Skills: Java, Python, R, C++, Html, CSS, JavaScript, PHP, SQL, Relational databases, Assembly language, Verilog HDL, Proteus, Arduino, WeBots, LabView, OOP, Algorithms, Django, Data Structure, MIP, LaTex, Bash/shell scripting, MatLab, GeoDa
Interests: Travelling, Cricket, Reading Historical Fictions
Languages: Bengali (Native), English (Proficient), Italian (Basic)